

ARAVIND RAM

CLOUD INFRASTRUCTURE ENGINEER

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PROFFESIONAL SUMMARY

Cloud Infrastructure Engineer with 2+ years designing and automating AWS infrastructure at scale. Specializes in Terraform IaC, EKS cluster provisioning, multi-account AWS architecture, and cost-optimized infrastructure design. Experienced operating petabyte-scale data infrastructure with strong reliability and cost engineering practices.

SKILLS

- Cloud (AWS): EKS, EC2, VPC design, IAM (roles, IRSA, policies), Route53, ACM, S3 (lifecycle, cross-region replication),
- ASG, Cost Explorer, Lambda, CloudWatch, CloudTrail
- IaC: Terraform (modules, remote state, workspaces), Terragrunt
- Containers & Orchestration: Kubernetes (EKS), Helm, ArgoCD, Docker
- Observability: Prometheus, Grafana, AlertManager, Elastic Stack, AWS Cost Explorer + Lambda pipelines
- Languages: Python, Bash, Go, Java, JavaScript
- Data Platforms: ClickHouse, MongoDB, DocumentDB, Kakfa, Redis, Cassandra

EXPERIENCE

[FRESHWORKS, CHENNAI](#)

2023- PRESENT

Cloud Infrastructure Engineer

- AWS Infrastructure Automation
 - Provisioned and maintained production AWS infrastructure for ClickHouse analytics platform (60 TB+, 200+ servers) entirely via Terraform
 - Designed reusable Terraform modules for EKS clusters, VPCs, IAM roles, Route53 zones, ACM certificates, S3 buckets, Auto Scaling Groups, Security Groups
 - Implemented tiered storage architecture: ClickHouse TTL policies move cold data from EBS to S3; local EBS caching for hot data — reduced storage costs substantially by 35%
 - S3 cross-region replication for Disaster Recovery: hourly ClickHouse backups replicated across regions; automated restore workflows; improved RTO from hours to minutes
- Infrastructure Cost Engineering
 - Built AWS Cost Explorer + Lambda aggregation pipeline for per-tenant infrastructure cost visibility
 - Delivered cost dashboards to engineering and leadership for infrastructure spend optimization
- Kubernetes on AWS
 - Deployed EKS clusters with Karpenter for node lifecycle management (on-demand + spot blending)
 - Configured IRSA for workload-level IAM, ALB Ingress Controller, ExternalDNS with Route53 automation
 - Managed ClickHouse platform lifecycle: upgrades, capacity planning, performance benchmarking across versions

- CI/CD & Automation
 - GitHub Actions pipelines: infrastructure plan/apply automation, ECR image sync, Helm validation
 - Python automation: tenant provisioning, manifest generation, ArgoCD deployment integration
 - Built remote data migration framework: ClickHouse Remote Functions for customer cluster onboarding; Route53
 - cutover procedures
 - Reliability
 - Designed and executed DR framework: automated backups, cross-region replication, snapshot recovery
 - Owned on-call for infrastructure incidents; built SOPCTL automation for faster incident resolution
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PROJECTS

- Infrastructure Cost Dashboard — AWS Cost Explorer + Lambda + custom dashboarding for per-tenant spend
- ClickHouse MCP Server — AI-assisted infra operational tooling
- MongoDB Migration — 16 TB DocumentDB → self-hosted MongoDB on Kubernetes

EDUCATION

Bachelor of Science (BS) Degree in Data Science and Applications

Indian Institute of technology, Madras 2023 - 2027

COURSEWORK

CKAD: Certified Kubernetes Application Developer

LANGUAGES

Tamil - Native

English - Fluent